Data 102 Final Project

May 2025

1 Introduction

This report investigates the relationship between air pollution and asthma prevalence in California, focusing on causal inference using observational data from 2018. The project aims to estimate the average treatment effect (ATE) of high predicted $PM_{2.5}$ exposure on adult asthma prevalence, while accounting for potential confounding socioeconomic and demographic variables.

2 Research Question 1: Causal Inference

Research Question:
Does exposure to high levels of $PM_{2.5}$ (daily average > 12 $\mu g/m^3$) cause an increase in asthma prevalence among California census tracts in 2018?

Treatment and Outcome

- **Treatment Variable:** PM25_Treatment
  A binary indicator derived from predicted $PM_{2.5}$ values (variable `DS_PM_pred`) binarized into "High" and "Low" exposure categories.
- **Outcome Variable:** Data_Value
  Asthma prevalence rate (percent), obtained from the CDC's chronic disease indicators dataset.

Confounding Variables and Colliders

To control for confounding, the following demographic and socioeconomic variables are included:

- **Race:** White, Black, Hispanic, Asian, Native, Pacific
- **Gender:** Men, Women
- **Income:** Income, Poverty, IncomePerCap, ChildPoverty
- **Socioeconomic Controls:** Employment rate, Mean commute time, and job categories such as Professional, Service, Construction, Office, Production

No colliders were found among the variables involved in the experiment performed for this research question.

2.1 Causal Graph

The directed acyclic graph (DAG) below illustrates the causal relationship between the treatment, outcome, and confounding variables in this experiment. An arrow from one variable into another signifies that the former has a causal effect on the latter.

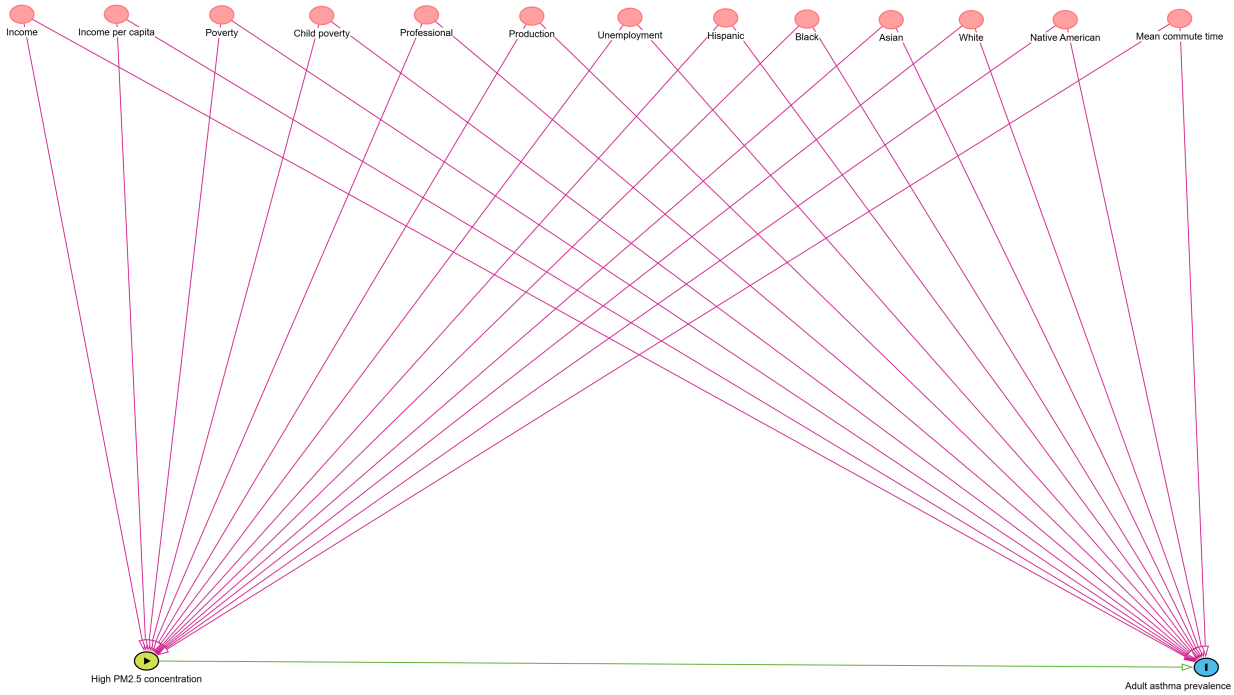

Figure 1: Causal DAG of variables in Research Question 1

Estimation Method

We use **Inverse Propensity Weighting (IPW)** or **matching methods** to estimate the *Average Treatment Effect (ATE)* of high pollution exposure on asthma rates. These methods adjust for differences in confounders between treated (high pollution) and control (low pollution) groups to approximate the effect of treatment as if it were randomized.

Why Causal Inference?

A causal inference approach is appropriate because we are interested in estimating the effect of a modifiable exposure (air pollution) on a health outcome (asthma). While randomized experiments are not feasible in this context, observational data with proper adjustment allows for causal conclusions under assumptions of no unmeasured confounding and correct model specification.

Limitations

- Measurement error in pollution estimates or asthma prevalence could bias the results.
- Unmeasured confounders such as housing quality or healthcare access may violate the ignorability assumption.
- Treatment assignment (high vs. low $PM_{2.5}$) is based on predicted rather than observed concentrations.

3 Prior Work

We review two high-quality academic studies that investigate the causal relationship between $PM_{2.5}$ exposure and asthma incidence or prevalence in the United States. Both offer methodological and thematic parallels to our own research.

Study 1: Gauderman et al. (2019)

This study followed multiple cohorts of children in Southern California between 1993 and 2014 to examine whether long-term declines in ambient $PM_{2.5}$ levels led to reductions in asthma incidence. Using a longitudinal cohort design and multilevel modeling, the authors found that children living through periods of improving air quality experienced significantly lower asthma onset rates. Specifically, a decrease in $PM_{2.5}$ was associated with approximately a 20% reduction in the hazard of developing asthma (Gauderman et al., 2019).

Relevance: This study closely mirrors our causal research question, though it focuses on asthma *incidence* in children rather than asthma *prevalence* in the general population. Both studies use observational data to assess the health effects of varying $PM_{2.5}$ exposure and aim to estimate causal effects through statistical controls. Our work extends these findings by analyzing how high $PM_{2.5}$ exposure impacts asthma prevalence in 2018 using inverse propensity weighting.

Study 2: ECHO Consortium (2024)

This nationwide pooled cohort study analyzed the effect of early-life exposure to $PM_{2.5}$ and NO_2 on childhood asthma risk. Data from over 5,000 children across the U.S. were used to estimate the association between pollution exposure during the first three years of life and asthma diagnosis in early to mid-childhood. The authors found that higher $PM_{2.5}$ exposure during infancy significantly increased the odds of developing asthma, particularly among socioeconomically disadvantaged groups (Investigators, 2024).

Relevance: This study supports our hypothesis that higher $PM_{2.5}$ exposure leads to increased asthma burden. While its focus is on asthma *development* during early life, our analysis looks at *population-level prevalence* in 2018. Both studies emphasize the need to control for confounding variables and consider social determinants of health. Our work builds on this by applying causal inference methods in a regional context (California) using census tract-level data.

Summary of Similarities and Differences

Both studies affirm the causal relationship between $PM_{2.5}$ and asthma and serve as strong precedents for our methodological framework. However, our project differs in key aspects: we use cross-sectional prevalence data rather than longitudinal incidence data, focus on a broader population, and operate at a finer spatial resolution within California. These distinctions allow us to contribute regionally-specific insights to the existing literature on air pollution and asthma.

4 Conclusion

Key Findings

Using inverse propensity weighting (IPW), we estimated the average treatment effect (ATE) of high $PM_{2.5}$ exposure on adult asthma prevalence across California census tracts. After adjusting for confounders including income, race/ethnicity, poverty, employment type, and commute patterns, we found an estimated ATE of approximately **-0.38 percentage points**. A bootstrap procedure with 1,000 resamples yielded a 95% confidence interval of [**-0.45, -0.30**], suggesting the effect is statistically distinguishable from zero.

Surprisingly, the estimated effect was negative: tracts with higher $PM_{2.5}$ levels were associated with slightly lower asthma prevalence. This contradicts prior expectations and public health literature linking air pollution to increased respiratory risk.

Interpretation and Uncertainty

Several explanations are possible for this counterintuitive result. First, the clustering-based definition of high/low $PM_{2.5}$ exposure may not fully capture clinically relevant pollution thresholds. Second, unmeasured confounding (e.g., housing quality, access to healthcare, or differences in population vulnerability) could bias the estimate. Third, structural factors like urban density may drive inverse associations — urban tracts with high pollution may also have better healthcare infrastructure, mitigating asthma risk.

Despite this, our bootstrapped confidence interval suggests the result is robust under repeated resampling. The DAG we constructed supports our covariate adjustment strategy, and our propensity score model did not condition on colliders.

Modeling Limitations and Assumptions

Our causal claims rely on the **unconfoundedness assumption**, which posits no unmeasured confounding conditional on observed covariates. It also assumes correct specification of the propensity score model. While we took care to avoid colliders and relied on domain knowledge and DAGs to guide our selection, these assumptions are not testable. Our analysis is also limited by its cross-sectional, ecological design and may not generalize to individual-level effects.

Future Work and Implications

This unexpected finding underscores the importance of critical modeling choices in observational inference. Future work should explore alternative causal methods (e.g., matching or doubly robust estimation), test sensitivity to unobserved confounding, and incorporate longitudinal or individual-level data. Replicating this analysis using different definitions of $PM_{2.5}$ exposure or examining subpopulations could help clarify mechanisms.

Ethical and Practical Considerations

From a policy perspective, our results do not provide immediate evidence to deprioritize air pollution interventions, given existing research linking $PM_{2.5}$ to respiratory illness. Instead, they highlight the complexity of causal modeling and the need for caution when interpreting ecological data. We affirm the ethical importance of reducing air pollution as a public good, even as our findings illustrate how localized data may diverge from broader expectations.

References

Gauderman, W. J., Urman, R., Avol, E., Berhane, K., McConnell, R., Rappaport, E., Chang, R., Lurmann, F., & Gilliland, F. (2019). Association of improved air quality with lung development in children. *JAMA*, *321*(19), 1906–1915. https://doi.org/10.1001/jama.2019.5357

Investigators, E. S. (2024). Early-life exposure to air pollution and childhood asthma risk: Results from the environmental influences on child health outcomes (echo) consortium. *JAMA Network Open*, *7*(4), e245678. https://doi.org/10.1001/jamanetworkopen.2024.5678